Topic 5: DataFrame

Learning Outcomes:
(a) Identify basic Pandas DataFrame properties and methods (shape, columns, index, info(), head() and tail())

Topic 5: DataFrame

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Why Pandas?

Easy to use
Efficient data structure
Powerful tools
Simple data merging
Pandas can cope with missing values
Simple column and row operations
You can use Pandas (with Python) in several domains
You can use Pandas with other libraries
Easier data munging
Data is flexible
Employees can complete plenty of work with less code

What is Pandas

- Derive form the term **Pan**el **Da**ta and also refers to **Python Data Analysis**
- Powerful Python library for handling and manipulating tables data or tabular data
- Pandas represent data in the form of - **Series** and **DataFrame**
- In summary, Pandas is a Python library used for data manipulation and analysis

Pandas Series and DataFrame

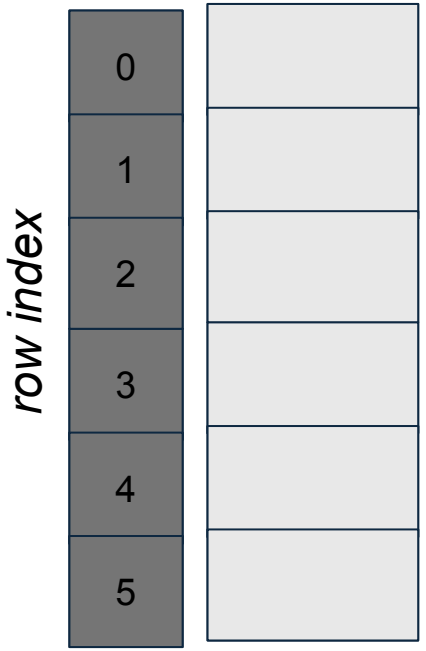
Pandas provides two types of classes for handling data:

1. **`Series`: a one-dimensional labeled array holding data of any type** such as integers, strings, Python objects etc.
2. **`DataFrame`**: a two-dimensional data structure that holds data like a two-dimensional array or a table with rows and columns.

Pandas Data Structure

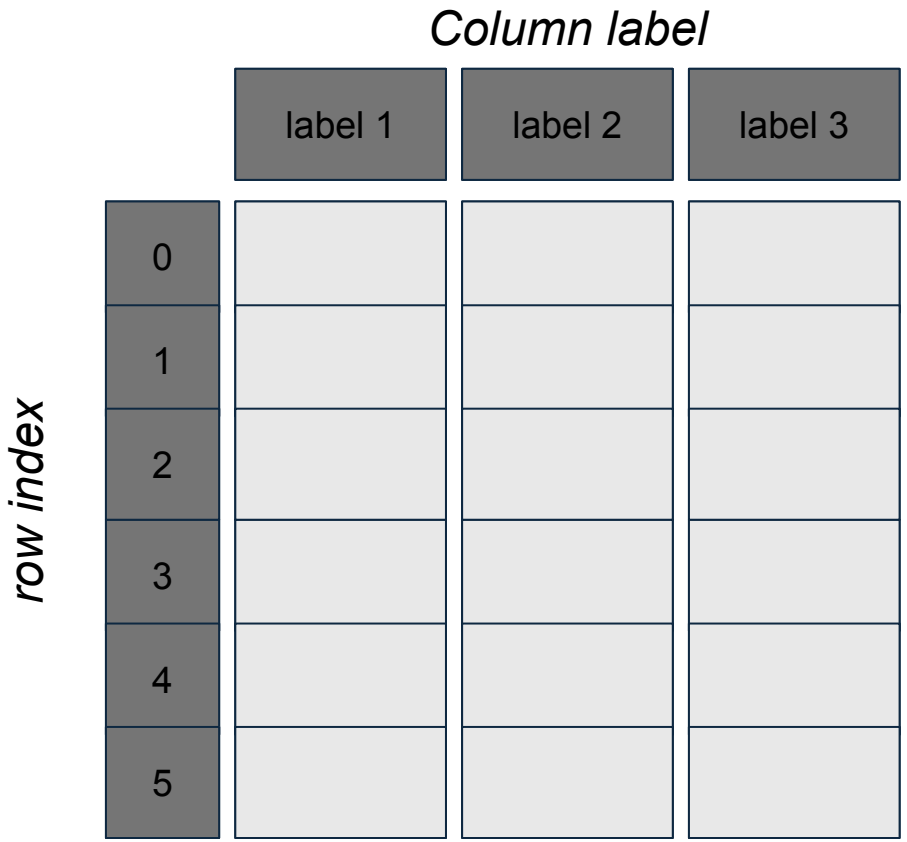
Series

Array-like object containing sequence of value
Similar to a column in table



DataFrame

Collection of series



Pandas Dataframe Overview

Series

	apples
0	3
1	2
2	0
3	1

+

Series

	oranges
0	0
1	3
2	7
3	2

=

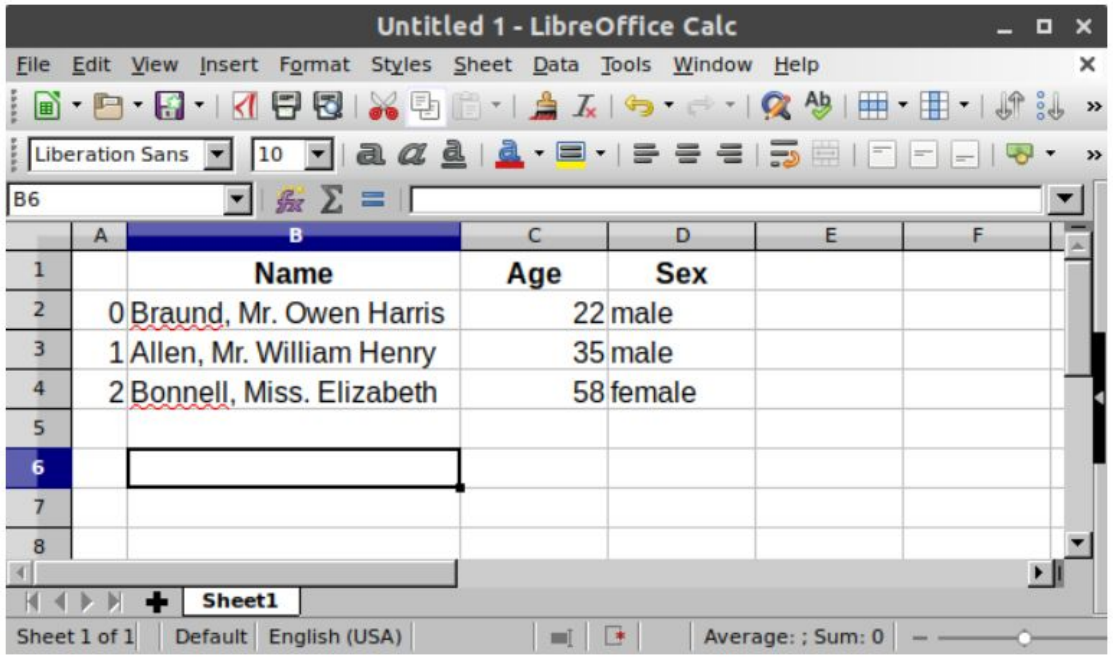
DataFrame

	apples	oranges
0	3	0
1	2	3
2	0	7
3	1	2

DataFrame is basically Pandas data table

DataFrame has similar structure to data representation in spreadsheet software

Spreadsheet



DataFrame

	Name	Age	Sex
0	Braund, Mr. Owen Harris	22	male
1	Allen, Mr. William Henry	35	male
2	Bonnell, Miss. Elizabeth	58	female

Data Type in Pandas

- A DataFrame is a 2-Dimensional data structure
- It can store data of different types including characters, **integers, floating point values, categorical data** and **more** in **columns.**

DataFrame

	Name	Age	Sex
0	Braund, Mr. Owen Harris	22	male
1	Allen, Mr. William Henry	35	male
2	Bonnell, Miss. Elizabeth	58	female

Data type

column names

```
Name    object
Age      int64
Sex     object
dtype: object
```

type

Data types in Pandas (Continue)

Data type	Usage
object	Text
int64	Integer numbers
float64	Floating point numbers
bool	True/False values
datetime64	Date and time values
timedelta[ns]	Differences between two datetimes
category	Finite list of text values

Getting Started with Pandas

Importing Pandas module

```
import pandas as pd
```

import statement

module name

alias

renaming **pandas** as **pd** as alias is considered as standard practice and use everytime pandas module is loaded into the program

Constructing Dataframe

To create a Pandas DataFrame, you can use Python to construct it from

- Lists or dictionaries,
- Import data from files like CSV or Excel,
- Connect to SQL databases,
- Fetch JSON data,
- Retrieve data from URLs, or even from
- Python dictionaries of Pandas Series

we will only focus on list and csv file

Constructing Dataframe (Cont')

- From List

```
data  =  [['Alice', 25], ['Bob', 30], ['Charlie', 35]]

df    =  pd.DataFrame(data,columns=[‘name’,’age’])
```

- From csv File

```
df=pd.read_csv(file_path)
```

Convert List to DataFrame

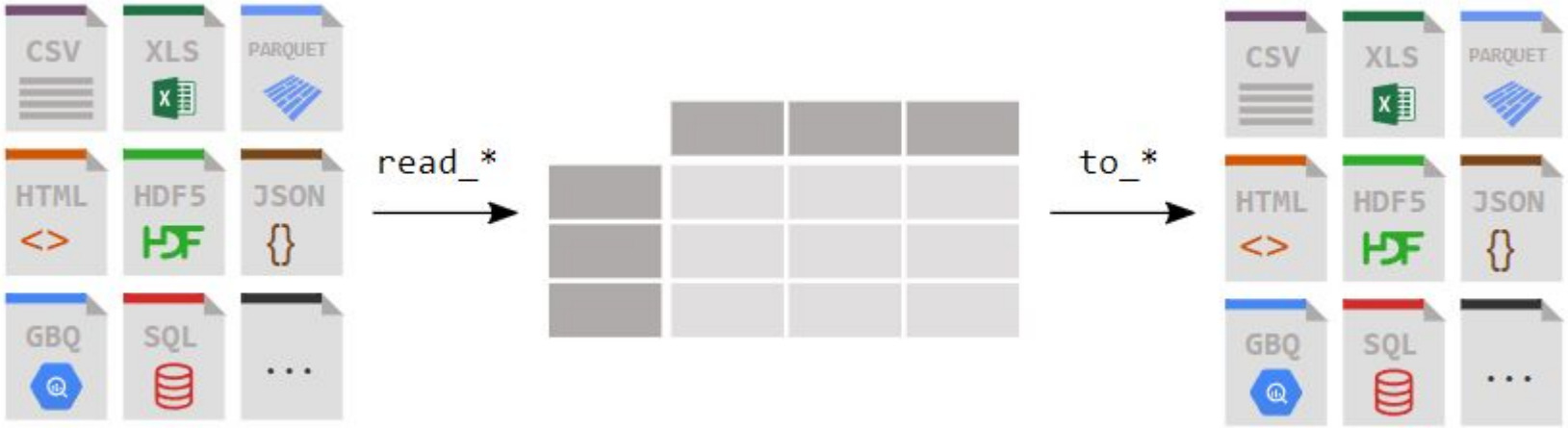
```
import pandas as pd

data = [
    ['Alice', 25, 'New York'],
    ['Bob', 30, 'Los Angeles'],
    ['Charlie', 35, 'Chicago']
]

df = pd.DataFrame(data, columns=['Name', 'Age', 'City'])
```

Output

```
      Name  Age         City
0    Alice   25     New York
1      Bob   30  Los Angeles
2  Charlie   35      Chicago
```

Load csv file into Pandas DataFrame

Pandas support many different data sources and file formats.



replace the * with the appropriate file format.

```
import pandas as pd
df = pd.read_csv('file_path')
```

Retrieve DataFrame Basic Information

To get a glimpse of the data, we can use the following methods

- df.shape #return tuple (row,col)
- df.columns #index column information
- df.index #row start index,stop index and step
- df.info() #summary of the DataFrame
- df.head() #Default to first 5 rows
- df.tail() #Default to last 5 rows

Dataframe from List

1. Shape of Dataframe

Return the number of rows and columns in the DataFrame

`df.shape`

```
import pandas as pd
```

```
data = [['Alice',25,'New York'],
        ['Bob',30,'Los Angeles'],
        ['Adam',35,'Chicago'],
        ['Tina',33,'Texas'],
        ['Violet',24,'Mexico'],
        ['Sam',37,'San Francisco']]
column_names = ['name','age','city']
```

```
# Creating Dataframe
df=pd.DataFrame(data,columns=column_names)
```

```
df
```

	name	age	city
0	Alice	25	New York
1	Bob	30	Los Angeles
2	Adam	35	Chicago
3	Tina	33	Texas
4	Violet	24	Mexico
5	Sam	37	San Francisco

‹t steps: Generate code with df View recommended plots

```
df.shape
```

```
(6, 3)
```

2. Get index in the Dataframe

Return start, end and steps in index

`df.index`

```
import pandas as pd
```

```
data = [['Alice',25,'New York'],
        ['Bob',30,'Los Angeles'],
        ['Adam',35,'Chicago'],
        ['Tina',33,'Texas'],
        ['Violet',24,'Mexico'],
        ['Sam',37,'San Francisco']]
column_names = ['name','age','city']
```

```
# Creating Dataframe
df=pd.DataFrame(data,columns=column_names)
```

```
df.index
```

```
RangeIndex(start=0, stop=6, step=1)
```

3. Get all columns in Dataframe

Return Index object containing the columns name

`df.columns`

```
import pandas as pd
```

```
data = [['Alice',25,'New York'],
        ['Bob',30,'Los Angeles'],
        ['Adam',35,'Chicago'],
        ['Tina',33,'Texas'],
        ['Violet',24,'Mexico'],
        ['Sam',37,'San Francisco']]
column_names = ['name','age','city']
```

```
# Creating Dataframe
df=pd.DataFrame(data,columns=column_names)
```

```
df.columns
```

```
Index(['name', 'age', 'city'], dtype='object')
```

4. Get Info of the Dataframe

Return index information, number of non-Null values in each columns, the data types and memory usage

```
df.info()
```

```python
import pandas as pd
```

```python
data = [['Alice',25,'New York'],
        ['Bob',30,'Los Angeles'],
        ['Adam',35,'Chicago'],
        ['Tina',33,'Texas'],
        ['Violet',24,'Mexico'],
        ['Sam',37,'San Francisco']]
column_names = ['name','age','city']
```

```python
# Creating Dataframe
df=pd.DataFrame(data,columns=column_names)
```

```python
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   name    6 non-null      object
 1   age     6 non-null      int64
 2   city    6 non-null      object
dtypes: int64(1), object(2)
memory usage: 272.0+ bytes
```

5a. Get First 5 Rows of the Dataframe

Default - Return first 5 rows

```
df.head()
```

```
import pandas as pd
```

```
data = [['Alice',25,'New York'],
        ['Bob',30,'Los Angeles'],
        ['Adam',35,'Chicago'],
        ['Tina',33,'Texas'],
        ['Violet',24,'Mexico'],
        ['Sam',37,'San Francisco']]
column_names = ['name','age','city']
```

```
# Creating Dataframe
df=pd.DataFrame(data,columns=column_names)
```

```
df.head()
```

	name	age	city
0	Alice	25	New York
1	Bob	30	Los Angeles
2	Adam	35	Chicago
3	Tina	33	Texas
4	Violet	24	Mexico

5b. Get First n rows of the Dataframe

Return first n rows

```
df.head(n)
```

```
import pandas as pd
```

```
data = [['Alice',25,'New York'],
        ['Bob',30,'Los Angeles'],
        ['Adam',35,'Chicago'],
        ['Tina',33,'Texas'],
        ['Violet',24,'Mexico'],
        ['Sam',37,'San Francisco']]
column_names = ['name','age','city']
```

```
# Creating Dataframe
df=pd.DataFrame(data,columns=column_names)
```

```
df.head(3)
```

	name	age	city
0	Alice	25	New York
1	Bob	30	Los Angeles
2	Adam	35	Chicago

6a. Get Last 5 Rows of the Dataframe

Default - Return last 5 rows

```
df.tail()
```

```
import pandas as pd
```

```
data = [['Alice',25,'New York'],
        ['Bob',30,'Los Angeles'],
        ['Adam',35,'Chicago'],
        ['Tina',33,'Texas'],
        ['Violet',24,'Mexico'],
        ['Sam',37,'San Francisco']]
column_names = ['name','age','city']
```

```
# Creating Dataframe
df=pd.DataFrame(data,columns=column_names)
```

```
df.tail()
```

	name	age	city
1	Bob	30	Los Angeles
2	Adam	35	Chicago
3	Tina	33	Texas
4	Violet	24	Mexico
5	Sam	37	San Francisco

6b. Get Last n Rows of the Dataframe

Return last n rows

```
df.tail(n)
```

```
import pandas as pd
```

```
data = [['Alice',25,'New York'],
        ['Bob',30,'Los Angeles'],
        ['Adam',35,'Chicago'],
        ['Tina',33,'Texas'],
        ['Violet',24,'Mexico'],
        ['Sam',37,'San Francisco']]
column_names = ['name','age','city']
```

```
# Creating Dataframe
df=pd.DataFrame(data,columns=column_names)
```

```
df.tail(2)
```

	name	age	city
4	Violet	24	Mexico
5	Sam	37	San Francisco

Dataframe from CSV files

#Reading CSV files

students.csv

	A	B	C
1	name	age	city
2	Alice	25	New York
3	Bob	30	Los Angeles
4	Adam	35	Chicago
5	Tina	33	Texas
6	Violet	24	Mexico
7	Sam	37	San Francisco

```
import pandas as pd
df = pd.read_csv("/content/drive/MyDrive/Colab Notebooks/students.csv")
```

1. Shape of Dataframe

Return the number of rows and columns in the DataFrame

`df.shape`

```
df.shape
```

```
(6, 3)
```

2. Get index in the Dataframe

Return start, end and steps in index

```
df.index
```

```
df.index
```

```
RangeIndex(start=0, stop=6, step=1)
```

3. Get all columns in Dataframe

Return Index object containing the columns name

```
df.columns
```

```
df.columns
```

```
Index(['name', 'age', 'city'], dtype='object')
```

4. Get Info of the Dataframe

Return index information, number of non-Null values in each columns, the data types and memory usage

```
df.info()
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 6 entries, 0 to 5
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   name    6 non-null      object
 1   age     6 non-null      int64
 2   city    6 non-null      object
dtypes: int64(1), object(2)
memory usage: 276.0+ bytes
```

5a. Get First 5 Rows of the Dataframe

Default - Return first 5 rows

```
df.head()
```

```
df.head()
```

	name	age	city
0	Alice	25	New York
1	Bob	30	Los Angeles
2	Adam	35	Chicago
3	Tina	33	Texas
4	Violet	24	Mexico

5b. Get First n rows of the Dataframe

Return first n rows

```
df.head(n)
```

```
df.head(3)
```

	name	age	city
0	Alice	25	New York
1	Bob	30	Los Angeles
2	Adam	35	Chicago

6a. Get Last 5 Rows of the Dataframe

Default - Return last 5 rows

```
df.tail()
```

```
df.tail()
```

	name	age	city
1	Bob	30	Los Angeles
2	Adam	35	Chicago
3	Tina	33	Texas
4	Violet	24	Mexico
5	Sam	37	San Francisco

6b. Get Last n Rows of the Dataframe

Return last n rows

```
df.tail(n)
```

```
df.tail(3)
```

	name	age	city
3	Tina	33	Texas
4	Violet	24	Mexico
5	Sam	37	San Francisco

Summary

- Learning to create Pandas DataFrame from list and CSV files
- Using Pandas DataFrame properties and methods to retrieve basic information about the DataFrame.